Cell Transmission Model for Mixed Traffic Flow with Connected and Autonomous Vehicles

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Abstract: The cell transmission model (CTM) is important for solving macroscopic traffic problems. Although some studies have been conducted for multiclass CTM, a simplified CTM for traffic flow mixed with connected and autonomous vehicles (CAVs) is still needed. This paper extends the homogenous CTM to form a mixed CTM under different CAV market penetration rates. To deal with this, the mixed triangular fundamental diagram is derived with different proportions of various vehicle types. The maximum capacity and backward wave speed of the mixed traffic flow are described by vehicle class proportions. Based on the mixed triangular fundamental diagram, the mixed CTM is proposed for different CAV penetration rates. An example application to validate the usefulness of the proposed mixed CTM is carried out. The mixed traffic flow in the application consists of cooperative adaptive cruise control (CACC), adaptive cruise control (ACC), and human vehicles with random vehicle orders. It is defined that CAVs travel under ACC when following a human vehicle without vehicle-to-vehicle (V2V) communication. Otherwise, the CACC function applies. The expected proportions of the CACC, ACC, and human vehicles in the random mixed traffic flow are described by the CAV penetration rate. Mixed CTM numerical experiments are conducted to evaluate the impacts of traffic accidents under different CAV penetration rates. Moreover, car-following simulations are performed to show consistency between macroscopic CTM numerical experiments and microscopic car-following simulations. The example application reveals that the proposed mixed CTM is simple and practical as applied to solve the traffic problem under different CAV penetration rates.
DOI: 10.1061/JTEPBS.0000238.

Author keywords: Connected and autonomous vehicle; Cell transmission model; Mixed traffic flow; Fundamental diagram.

Introduction

Connected and autonomous vehicles (CAVs) are promoting the development of traffic systems. Generally speaking, CAVs have two types of operation. Under adaptive cruise control (ACC), onboard sensors detect distance gaps and speed differences between the CAV and a preceding vehicle, which are usually used to optimize CAV acceleration/deceleration. In contrast, vehicle-to-vehicle (V2V) communication is helpful for obtaining acceleration information from a preceding vehicle. In this situation, CAVs travel under cooperative adaptive cruise control (CACC).

Research in modeling CAV traffic flow has become a hot topic in recent years (Mahmassani 2016). Many studies (Ardakani and Yang 2017; Gong et al. 2016; Gong and Du 2018; Jia and Ngoduy 2016a, b; Qin and Wang 2018; van Arem et al. 2006; Wang et al. 2018) have been conducted on microscopic CAV models—a car-following model, for example. However, the research is not well balanced because the literature on macroscopic CAV models is relatively sparse. Macroscopic models are essentially applied in real-time traffic management because real-time applications require computational efficiency (Tiaprasert et al. 2017). The Lighthill-Whitham-Richards (LWR) model (Lighthill and Whitham 1955; Richards 1956), based on kinematic wave theory, is a first-order macroscopic model. Some studies extended this model to a high-order macroscopic model for capturing more complex traffic phenomena (Hoogendoorn and Bovy 2000; Jin 2017; Ngoduy 2013). Jin (2016) studied the equivalence between a macroscopic continuum model and a microscopic car-following model. As one of the most well-known discrete approximation of the LWR model, Daganzo (1994, 1995) presented the cell transmission model (CTM). Since it requires less computation, CTM has been widely used to solve many traffic problems (Han et al. 2017; Lo et al. 2001). This paper focuses on the CTM for mixed CAV traffic flow.

For traditional mixed traffic flow, such as car-truck vehicular flow, the original CTM (Daganzo 1994, 1995) has been extended to a multiclass CTM (Tuerprasert and Aswakul 2010). Lane-changing behavior has also been explained in some studies (Carey et al. 2015; Laval and Daganzo 2006). We refer readers to Qian et al. (2017) for a more detailed understanding of CTM extensions. What mainly concerns us here is a CTM of mixed traffic flow consisting of CAVs and human vehicles. Levin and Boyles (2016a) presented a CTM to evaluate traffic flow behaviors in dynamic lane reversal with CAVs. A multiclass CTM was also proposed that takes into consideration the different reaction times of CAVs and human vehicles (Levin and Boyles 2016b). Tiaprasert et al. (2017) proposed a multiclass CTM enhanced with overtaking, lane-changing, and first-in first-out properties. However, most studies from the literature make the original CTM complicated (Daganzo 1994) in an effort to capture more accurate vehicle behaviors, such as lane-changing and overtaking. From a macroscopic perspective, what is of concern is vehicular flow evolution as a whole, regardless of local vehicle behaviors, under different CAV market penetration rates, in mixed traffic flow. Therefore, a simple and practical CTM for the mixed CAVs flow is still needed. This is the subject of this paper.

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Note. This manuscript was submitted on April 18, 2018; approved on October 22, 2018; published online on February 27, 2019. Discussion period open until July 27, 2019; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Transportation Engineering, Part A: Systems***, © ASCE, ISSN 2473-2907.

A trapezoidal/triangular fundamental diagram (Newell 1993) is essential in a CTM. Moreover, a mixed triangular fundamental diagram can describe the maximum capacity and backward wave speed of mixed traffic flow under various CAV penetration rates. Levin and Boyles (2016a, b) studied capacity and backward wave speed in designing a multiclass CTM without lane-changing and overtaking behaviors. However, they presented it from the perspective of a car-following model incorporating driver reaction time. To the best of our knowledge, little research has been conducted on a CTM of mixed CAV-human traffic flow from the perspective of a general triangular fundamental diagram with different CAV penetration rates. Therefore, this paper deals with this research gap by providing another modeling idea for a CTM of mixed CAV flow.

The organization of this paper is as follows. Mixed triangular fundamental diagrams are first derived for different proportions of vehicle types. Then the mixed CTM is proposed and its usefulness is validated in an example application. Finally, some conclusions are offered.

Mixed Triangular Fundamental Diagram

A CTM assumes a trapezoidal/triangular fundamental diagram (Newell 1993) for traffic flow. This section presents a triangular fundamental diagram of mixed flow with CAVs. The diagram requires a linear relationship between spacing and speed at equilibrium for both homogenous and mixed traffic flow. Equilibrium spacing is usually derived as a function of equilibrium speed as follows:

$$s_m = vT_m + d_m \tag{1}$$

where s_m = spacing of vehicle class m; v = equilibrium speed; T_m = reaction time of human vehicles or the time gap of CAVs belonging to class m; and d_m = jam spacing, usually including vehicle length.

Eq. (1) describes the desired spacing of an individual vehicle depending on its vehicle class and traffic speed. It can be seen that each individual vehicle's spacing is determined by its class and by its traffic speed, regardless of vehicle order.

In mixed traffic flow, vehicle speeds are the same at equilibrium, while individual vehicle spacing may vary. The subscript m of Eq. (1) stands for the vehicle class in terms of T_m and d_m, which includes different vehicle types and heterogeneous characteristics of individual vehicles. Assume that there are M classes in the mixed traffic flow. Then the density of the mixed flow is calculated as follows:

$$k = \frac{1}{\sum_{m=1}^{M} p_m(vT_m + d_m)} \tag{2}$$

where p_m = proportion of class m in the mixed traffic flow.

From Eq. (2), it can be seen that the density of mixed traffic flow is calculated based on equilibrium speed. Equilibrium speed is considered to be constant for each vehicle class in mixed traffic flow (Ge and Orosz 2014; Jia et al. 2018), which means that all vehicles have the same equilibrium speed under a certain traffic condition. In order to calculate the q-k relation, we make an adjustment for Eq. (2), which means that speed v can be described as a function of density k

$$v = \frac{1}{k\sum_{m=1}^{M}(p_m T_m)} - \frac{\sum_{m=1}^{M}(p_m d_m)}{\sum_{m=1}^{M}(p_m T_m)} \tag{3}$$

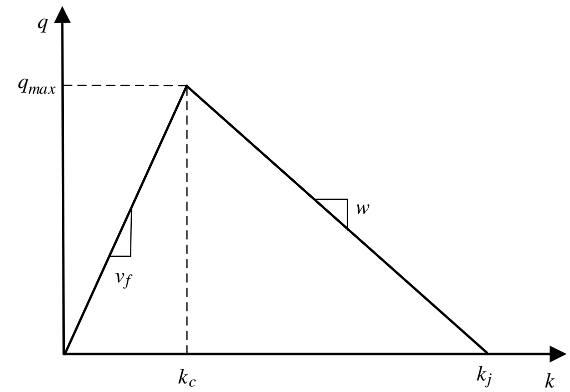

Fig. 1. Mixed triangular fundamental diagram with different p_m.

Therefore, the q-k relation of the mixed fundamental diagram can be obtained as

$$q = \frac{1}{\sum_{m=1}^{M}(p_m T_m)} - \frac{\sum_{m=1}^{M}(p_m d_m)}{\sum_{m=1}^{M}(p_m T_m)}k \tag{4}$$

The q-k relation in Eq. (4) describes the congested state of the fundamental diagram, while the free flow state can be added to obtain a triangular fundamental diagram, as illustrated in Fig. 1.

In Fig. 1, v_f is the free-flow speed, $q_{\max}$ is the maximum flow, k_c is the critical density at $q_{\max}$, k_j is the jam density, and w is the backward wave speed. For the triangular fundamental diagram, the critical density k_c is related to the free-flow speed v_f. Substituting $v = v_f$ into Eq. (2) to calculate k_c, we obtain

$$k_c = \frac{1}{\sum_{m=1}^{M} p_m(v_f T_m + d_m)} \tag{5}$$

Then the maximum flow $q_{\max}$ is

$$q_{\max} = \frac{v_f}{\sum_{m=1}^{M} p_m(v_f T_m + d_m)} \tag{6}$$

The jam density k_j can be calculated when v is equal to zero in Eq. (2), which yields

$$k_j = \frac{1}{\sum_{m=1}^{M}(p_m d_m)} \tag{7}$$

The backward wave speed is the slope of the triangular fundamental diagram. Based on Eq. (4), we have

$$w = -\frac{\sum_{m=1}^{M}(p_m d_m)}{\sum_{m=1}^{M}(p_m T_m)} \tag{8}$$

Eq. (8) shows that the properties of the mixed triangular fundamental diagram, such as maximum capacity $q_{\max}$ and backward wave speed w, are related to the proportion p_m of specific class m in the mixed traffic flow.

Mixed Cell Transmission Model

Here the mixed triangular fundamental diagram just derived is used to form the CTM of mixed traffic flow. Like Daganzo (1994), we discretize the time into steps Δt and the road into cells Δx, so $\Delta x = \Delta t \times v_f$ should be satisfied. The cells are labeled i, which is larger when the cell is located downstream. The mixed triangular fundamental diagram in Eq. (4) assumes that all vehicles travel at the same speed. Although some studies have focused on different vehicle speeds in the CTM (Tiaprasert et al. 2017; Tuerprasert and Aswakul 2010), assuming the same speed is reasonable because a vehicle might match the speed of surrounding vehicles, especially in the crowded state (Levin and Boyles 2016a, b). Traffic speed v is determined by the free-flow speed, capacity, and backward wave speed. According to Fig. 1, we can calculate v as follows:

$$v = \min\left\{v_f, \frac{q_{\max}}{k}, \frac{-w(k_j - k)}{k}\right\} \tag{9}$$

where k = mixed flow density.

Substituting Eqs. (6)–(8) into Eq. (9) yields

$$v = \min\left\{v_f, \frac{v_f}{k\sum_{m=1}^{M} p_m(v_f T_m + d_m)}, \frac{\sum_{m=1}^{M}(p_m d_m)}{\sum_{m=1}^{M}(p_m T_m)} \times \left(\frac{1}{k\sum_{m=1}^{M}(p_m d_m)} - 1\right)\right\} \tag{10}$$

Based on CTM theory (Daganzo 1994), cell transition flow can be calculated as follows:

$$y_i(t) = \min\left\{n_{i-1}(t), Q_{\max}, \frac{-w}{v_f}[N - n_i(t)]\right\} \tag{11}$$

where $y_i(t)$ = vehicles entering from cell $i-1$ to cell i at time t; n_{i-1} = vehicles in cell $i-1$ at time t; n_i = vehicles in cell i at time t; $Q_{\max}$ = maximum capacity limitation; and N = maximum vehicles that can fit in each cell.

Based on the mixed triangular fundamental diagram derived in Eq. (4), we have

$$Q_{\max} = q_{\max}\Delta t = \frac{\Delta x}{\sum_{m=1}^{M} p_m(v_f T_m + d_m)} \tag{12}$$

and

$$N = k_j \Delta x = \frac{\Delta x}{\sum_{m=1}^{M}(p_m d_m)} \tag{13}$$

Then the cell occupancy update is described as

$$n_i(t + \Delta t) = n_i(t) + y_i(t) - y_{i+1}(t) \tag{14}$$

Based on Eq. (14), we can obtain vehicles in each cell over time and then calculate density, speed, and flow over time for mixed traffic flow at different proportions p_m of vehicle class m. The proposed CTM assumes that class-specific density is uniformly distributed throughout the cells, which extends the assumption of a single-class CTM (Levin and Boyles 2016a). Levin and Boyles (2016b) considered Newell's car-following model (Newell 2002) to start the multiclass CTM. Newell's model is directly related to the triangular fundamental diagram, which is the linkage between this paper and Levin and Boyles (2016b).

Application

We consider a specific scenario of mixed traffic flow in order to perform an example application using the proposed mixed CTM. In addition, the corresponding car-following models are used for microscopic simulations, which are shown to be consistent with macroscopic CTM numerical experiments.

Mixed Flow Scenario

As mentioned previously, CAVs monitor a preceding vehicle's acceleration with V2V communication under CACC, but only detect the distance gap and speed difference to the lead vehicle using onboard sensors under ACC. Moreover, the time gap of CAVs under CACC can be smaller than that under ACC, which has benefits in capacity. CAVs are able to travel under CACC, but CACC cannot work without V2V communication. Therefore, the mixed traffic flow in the application is defined as follows: (1) when traffic flow mixes CAVs and human vehicles in random vehicle order, CAVs have V2V communication equipment but human vehicles do not; (2) when CAVs follow human vehicles, they have ACC; otherwise, have CACC.

To avoid confusion, CAVs under CACC and ACC are called CACC vehicles and ACC vehicles, respectively. Let the symbol p be the market penetration rate of CAVs in the random mixed traffic flow. From a mathematical expectation perspective, the probability that one CAV follows one human vehicle is the product of their proportions—namely, $p(1-p)$. This means that the expected ACC proportion is $p(1-p)$. Then the expected CACC proportion equals the proportion of CAVs minus that of ACC vehicles—namely, $p - p(1-p) = p^2$. Therefore, the expected proportions of CACC, ACC, and human vehicles can be described by the market penetration rate p of CAVs, as follows:

$$\begin{aligned} p_c &= p^2 \\ p_a &= p(1-p) \\ p_h &= 1-p \end{aligned} \tag{15}$$

where p_c, p_a, and p_h = expected proportions of CACC, ACC, and human vehicles, respectively.

Without loss of generality, Eq. (15) can also describe homogenous human flow if $p = 0$, while homogenous CACC flow is described if $p = 1$.

Macroscopic CTM Numerical Experiments

CTM is useful for evaluating the macroscopic dynamics of traffic flow with less computation. The mixed traffic flow defined in this paper is focused on evaluating flow evolution using the proposed mixed CTM. The mixed CTM numerical experiments deal with the expected situations of vehicle proportions described by the CAV penetration rate p in Eq. (15).

Accidents are a common traffic phenomenon and often cause queues during cleanup. If an accident occurs on a freeway, a period of time is needed for cleanup, during which vehicles upstream of the accident location form a queue. Normal operation is recovered after cleanup. Our concern here is impacts on the queue upstream under different values of p in mixed traffic flow. As an example, assume that arrival flow is fixed at 1,500 veh/h in mixed traffic flow. Fifteen minutes is needed for cleanup, during which a queue is gradually formed. Following cleanup, access to the freeway is again available and the queue begins to dissipate until normal operation is achieved.

According to Shladover et al. (2018), CACC vehicles can reduce the time gap to 0.6 s, while the lowest time gap for ACC vehicles is usually 1.1 s. Therefore, we consider $T_m = T_c = 0.6$ s for CACC vehicles in Eq. (1), in which the subscript c denotes

J. Transp. Eng., Part A: Systems, 2019, 145(5): 04019014

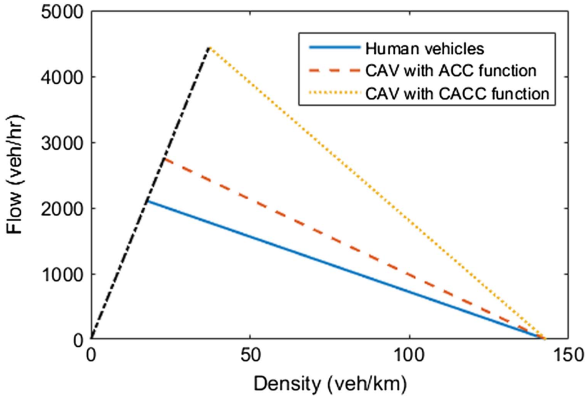

Fig. 2. Homogenous triangular fundamental diagram.

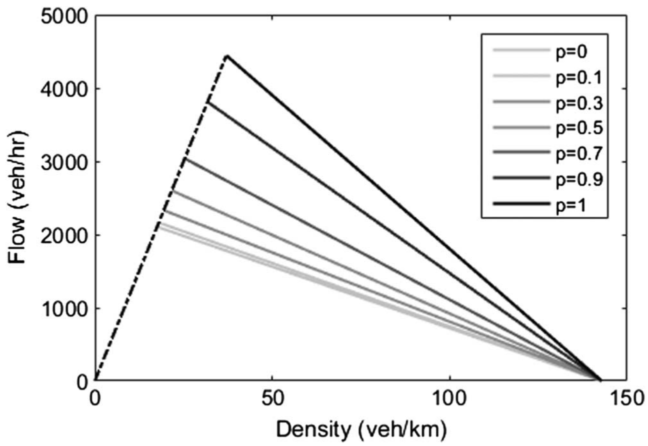

Fig. 3. Mixed triangular fundamental diagram for different CAV rates p.

the CACC vehicle class. Moreover, $T_m = T_a = 1.1$ s is considered in Eq. (1) for the ACC vehicle class. In the case of human vehicles, $T_m = T_h = 1.5$ s is considered based on previous studies (Ahn et al. 2004; Johansson and Rumar 2016; Levin and Boyles 2016b). The same d_m in Eq. (1) is considered for the three types of vehicles, set as 7 m as is usual (Kesting et al. 2008; Milanés and Shladover 2014). We substitute these values into Eq. (4) and let subscript m in Eq. (4) be c, a, h, respectively, to calculate the triangular fundamental diagram of homogenous CACC, ACC, and human flow, respectively. The calculation results are shown in Fig. 2. Moreover, we calculate the mixed triangular fundamental diagram under different CAV penetration rates p by substituting Eq. (15) into Eq. (4) as is shown in Fig. 3, which indicates that capacity gradually increases with the increase in p, because a smaller time gap results in higher capacity (Shladover et al. 2018).

In addition, cell length $\Delta x = 100$ m, discretized time step $\Delta t = 3$ s, and free-flow speed $v_f = 120$ km/h in the CTM numerical experiments, satisfying $\Delta x = \Delta t \times v_f$. Based on the mixed triangular fundamental diagram, we can calculate values for maximum capacity and vehicles for each cell under different values of p, following Eqs. (12) and (13). Then the cell occupancy update for mixed traffic flow can be calculated based on Eqs. (11) and (14).

The impact results for the traffic accident using macroscopic CTM numerical experiments are shown in Fig. 4. The grayscale bar in Fig. 4 denotes traffic flow speeds in cells over time. At the initial time, the vehicle speed is the free-flow speed under the arrival flow of 1,500 veh/h for mixed traffic flow in the triangular fundamental diagram. Moreover, the accident occurs at 20 km, where vehicle speed becomes zero during the 15-min cleanup time. Then Fig. 4 shows the queue caused by the traffic accident over space and time. It can be seen that queue length and impact time gradually decrease with the increase in p. More important, the mixed CTM proposed in this paper can deal with the traffic problem in terms of mixed CACC–ACC–human vehicular flow with different CAV market penetration rates p. Moreover, compared with the original CTM (Daganzo 1994), computation of the proposed mixed CTM does not increase because we use a mixed triangular fundamental diagram to replace the homogenous diagram in the original CTM (Daganzo 1994).

Microscopic Car-Following Simulations

Important in the traffic flow problem is analysis of the consistency between macroscopic results and microscopic simulations. Therefore, this section performs car-following simulations and compares their results with the those for the macroscopic mixed CTM numerical experiments. We use a linear car-following model for human vehicles and a model with a constant time gap for CACC/ACC vehicles because these models are consistent with the triangular fundamental diagram used in CTM.

In the case of human vehicles, Newell's lower order model (Newell 2002) is employed. The model equation can be written as follows:

$$x_n(t + T_h) = x_{n-1}(t) - d_h \tag{16}$$

where $x_{n-1}(t)$ = position of the preceding vehicle $n-1$ at time t; $x_n(t + T_h)$ = position of follower n after time T_h.

For consistency with the setting in the macroscopic mixed CTM numerical experiments, $T_h = 1.5$ s and $d_h = 7$ m in the car-following simulations.

The CACC car-following model with a constant time gap is written as

$$\ddot{x}_n(t) = k_0\ddot{x}_{n-1}(t) + k_1(x_{n-1}(t) - x_n(t) - v_n(t)T_c - d_c) + k_2(v_{n-1}(t) - v_n(t)) \tag{17}$$

where x, v, and $\ddot{x}$ = vehicle position, speed, and acceleration, respectively.

The same setting for T_c and d_c is satisfied—namely, $T_c = 0.6$ s and $d_c = 7$ m. Based on previous studies (Talebpour and Mahmassani 2016; van Arem et al. 2006), $k_0 = 1.0$, $k_1 = 0.1\ \mathrm{s}^{-2}$, and $k_2 = 0.58\ \mathrm{s}^{-1}$.

Without V2V communication, CAVs travel under ACC to respond to the distance gap and speed difference represented by the preceding vehicle. Then the ACC car-following model with constant time gap is written as follows:

$$\ddot{x}_n(t) = k_1[x_{n-1}(t) - x_n(t) - v_n(t)T_a - d_a] + k_2[v_{n-1}(t) - v_n(t)] \tag{18}$$

where $T_a = 1.1$ s, $d_a = 7$ m, and k_1 and $k_2 = k_1$ and k_2 in Eq. (17) in the car-following simulations.

The same traffic conditions for the traffic accident in the macroscopic CTM experiments are considered in the microscopic

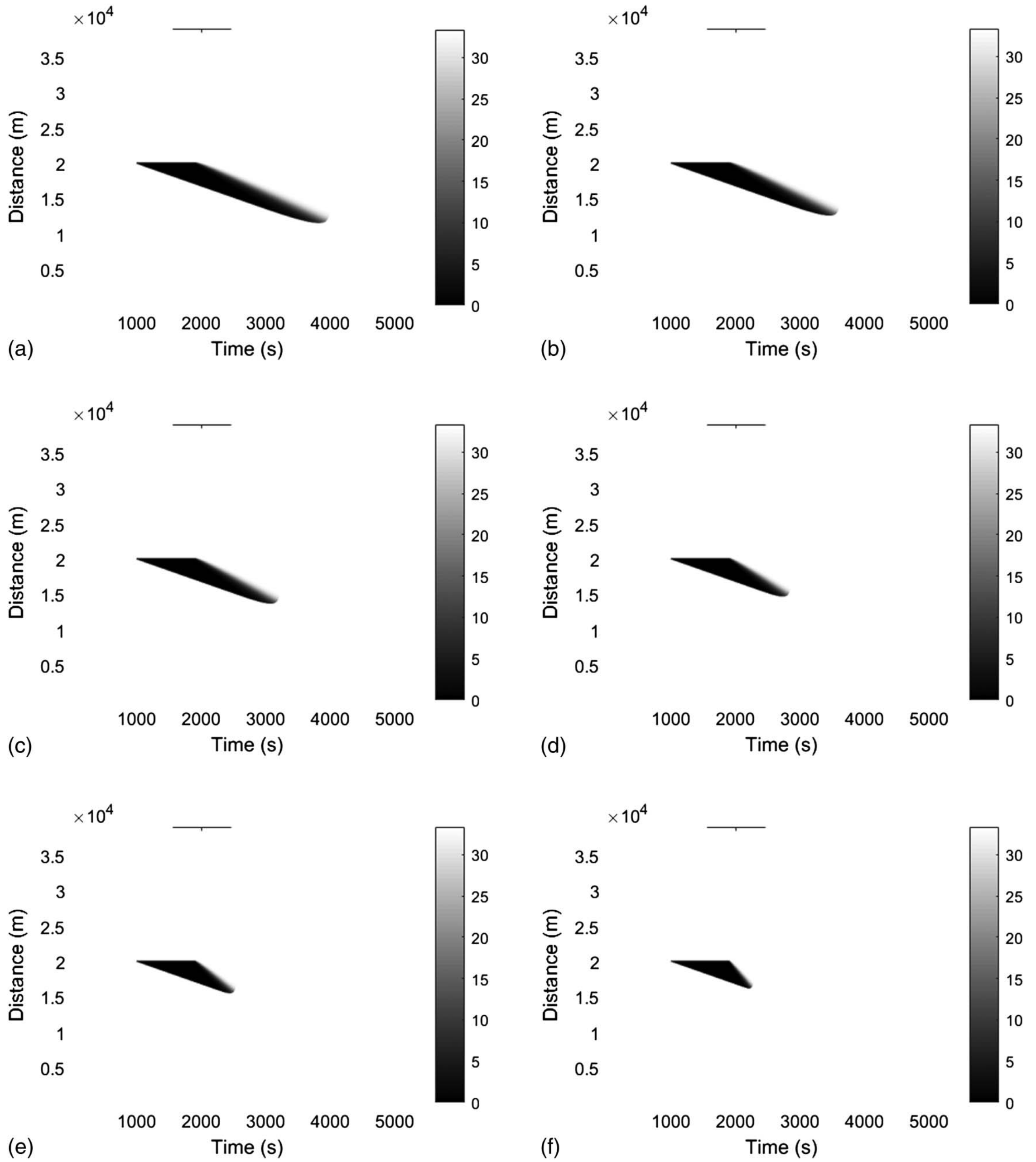

Fig. 4. Impact results for traffic accident in macroscopic mixed CTM numerical experiment: (a) $p = 0$; (b) $p = 0.2$; (c) $p = 0.4$; (d) $p = 0.6$; (e) $p = 0.8$; and (f) $p = 1$.

simulations. The microscopic simulations are performed in MATLAB using the three car-following models under different CAV penetration rates p. The randomness of vehicle order in the mixed traffic flow is considered for each simulation. Then the CAVs randomly become CACC or ACC depending on the preceding vehicle type in each case, based on the mixed traffic flow definition in this paper. The car-following simulation step time is 0.1 s, the maximum acceleration is 4 $\mathrm{m/s^2}$, and the emergency deceleration is -6 $\mathrm{m/s^2}$ (Ni et al. 2016). Heat maps of vehicle speed over all time and space are calculated, in which the time interval is 1 s and the space interval is 100 m, as shown in Fig. 5. Because our concern is the impacts of the traffic accident on flow dynamics, only the vehicles that travel upstream of the traffic accident location, when the accident occurs, are considered in calculating the simulation results in Fig. 5. This means that only vehicles affected by the accident are considered. A comparison of Figs. 4 and 5 shows that the impact results for the macroscopic numerical experiments with the proposed mixed CTM are almost consistent with those of the microscopic car-following simulations.

In order to illustrate wave speed changes for each vehicle, we provide time-space vehicle simulation trajectories based on car-following simulations. Fig. 6 shows the wave speed changes for vehicles with respect to the simulation time. Specifically, it shows that vehicles slow and finally stop in front of the accident location because of the cleanup. During this time, the queue appears. After that, vehicles accelerate and traffic finally returns to normal operation. We should also mention that the vehicles in Fig. 6 are sampled at a 10-vehicle interval in order to better illustrate the features of individual vehicles. The results of the CTM numerical experiments and car-following simulations reveal the usefulness of

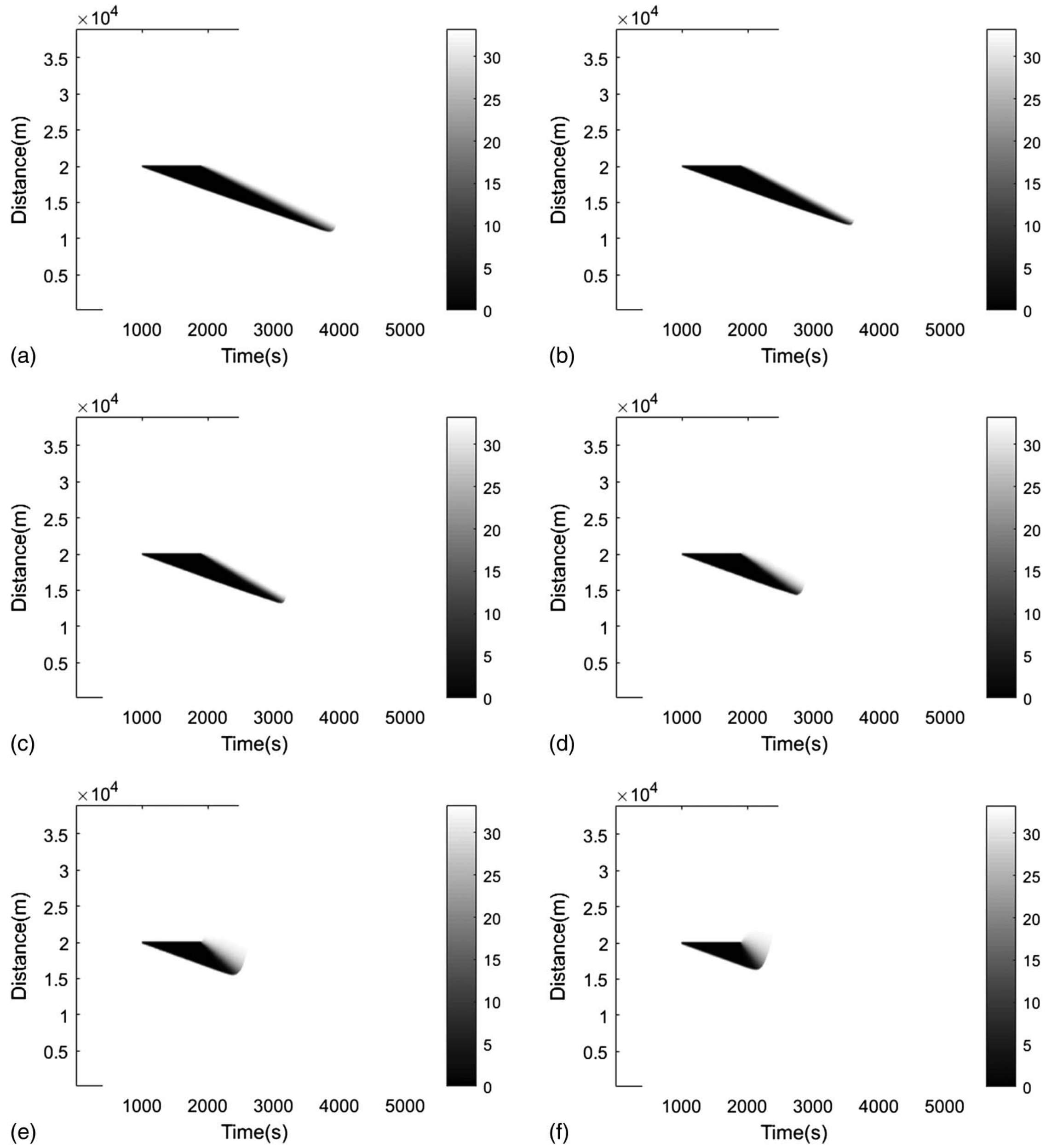

Fig. 5. Heat maps of speed over time and space in microscopic car-following simulations: (a) $p = 0$; (b) $p = 0.2$; (c) $p = 0.4$; (d) $p = 0.6$; (e) $p = 0.8$; and (f) $p = 1$.

the mixed CTM proposed in this paper for solving the traffic accident problem, from a macroscopic perspective, under different CAV penetration rates.

Conclusions

Some studies (e.g., Qian et al. 2017; Tiaprasert et al. 2017) have investigated multiclass CTMs taking into consideration complex vehicle behaviors such as lane changing and overtaking. However, a macroscopic perspective takes into consideration vehicular flow evolution as a whole, regardless of local vehicle behavior. Therefore, it needs a simple and practical CTM for mixed traffic flow with different CAV penetration rates. This paper describes a proposed simplified mixed CTM that extends the original CTM (Daganzo 1994). The mixed CTM is based on the mixed triangular fundamental diagram, which defines the proportions of different vehicle types. It deals with various scenarios of mixed traffic flow with different proportions of CAV and human vehicles.

To validate the usefulness of the proposed mixed CTM, an example application was conducted to solve traffic accident problem. The mixed traffic flow in the application was defined as random mixed CACC–ACC–human vehicular flow determined by the CAV market penetration rate. The mixed CTM numerical experiments provided macroscopic impacts of traffic accidents on a vehicle queue over space and time for different CAV penetration rates. Results show that the queue created by a traffic accident decreases with increasing CAV penetration rate. Moreover, the microscopic car-following simulations showed consistency with results for the macroscopic CTM experiments. As illustrated by the example application, the proposed mixed CTM is a simple and practical way to solve traffic problems with less computation.

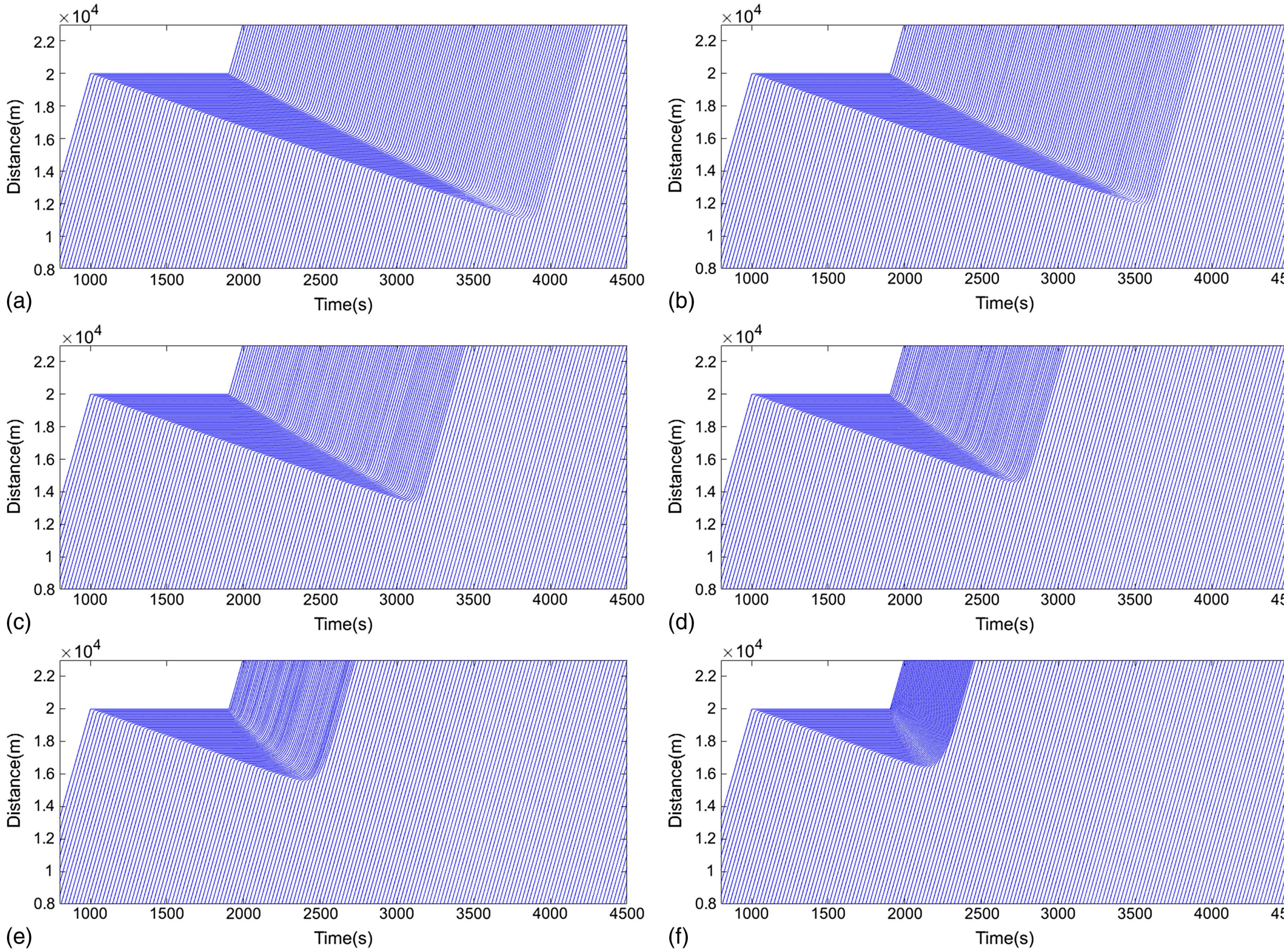

Fig. 6. Simulation trajectories for vehicles over time and space in microscopic car-following simulations: (a) $p = 0$; (b) $p = 0.2$; (c) $p = 0.4$; (d) $p = 0.6$; (e) $p = 0.8$; and (f) $p = 1$.

We are aware that the proposed mixed CTM has limitations. For example, it cannot track traffic flow dynamics accurately if the proportion of each vehicle class in each cell fluctuates greatly, because class-specific density is assumed to be uniformly distributed throughout the cells (Levin and Boyles 2016a). Moreover, the proposed mixed CTM should be further validated using available experimental data given that, at present, large-scale experimental CAV data are not available. Only small-scale experiments with several CAVs have been conducted; however, CAV equilibrium spacing-speed can be determined based on them and then the CTM proposed in this paper can be used to analyze realistic dynamics of mixed CAV flows.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 51478113 and 51878161); the Scientific Research Foundation of the Graduate School of Southeast University (Grant No. YBJJ1792), and the Fundamental Research Funds for the Central Universities and the Postgraduate Research & Practice Innovation Program of Jiangsu Province (Grant No. KYCX17_0146).

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